

College

Gao

2024-04-17

```
## -- Attaching packages ----- tidymodels 1.1.1 --
```

```
## v broom      1.0.5      v recipes      1.0.10
## v dials      1.2.1      v rsample      1.2.0
## v dplyr      1.1.4      v tibble      3.2.1
## v ggplot2    3.5.0      v tidyr       1.3.1
## v infer      1.0.6      v tune        1.1.2
## v modeldata  1.3.0      v workflows   1.1.4
## v parsnip    1.2.0      v workflowsets 1.0.1
## v purrr      1.0.2      v yardstick   1.3.0
```

```
## -- Conflicts ----- tidymodels_conflicts() --
```

```
## x purrr::discard() masks scales::discard()
## x dplyr::filter()   masks stats::filter()
## x dplyr::lag()      masks stats::lag()
## x recipes::step()   masks stats::step()
## * Use tidymodels_prefer() to resolve common conflicts.
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v forcats  1.0.0      v readr      2.1.5
## v lubridate 1.9.3      v stringr    1.5.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x readr::col_factor() masks scales::col_factor()
## x purrr::discard()    masks scales::discard()
## x dplyr::filter()     masks stats::filter()
## x stringr::fixed()    masks recipes::fixed()
## x dplyr::lag()        masks stats::lag()
## x readr::spec()       masks yardstick::spec()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
## Loading required package: rpart
```

```
##
```

```
##
```

```
## Attaching package: 'rpart'
```

```
##
```

```
##
```

```
## The following object is masked from 'package:dials':
```

```
##
```

```
##      prune
```

```
##
```

```
##
```

```
## Loading required package: lattice
```

```
##
```

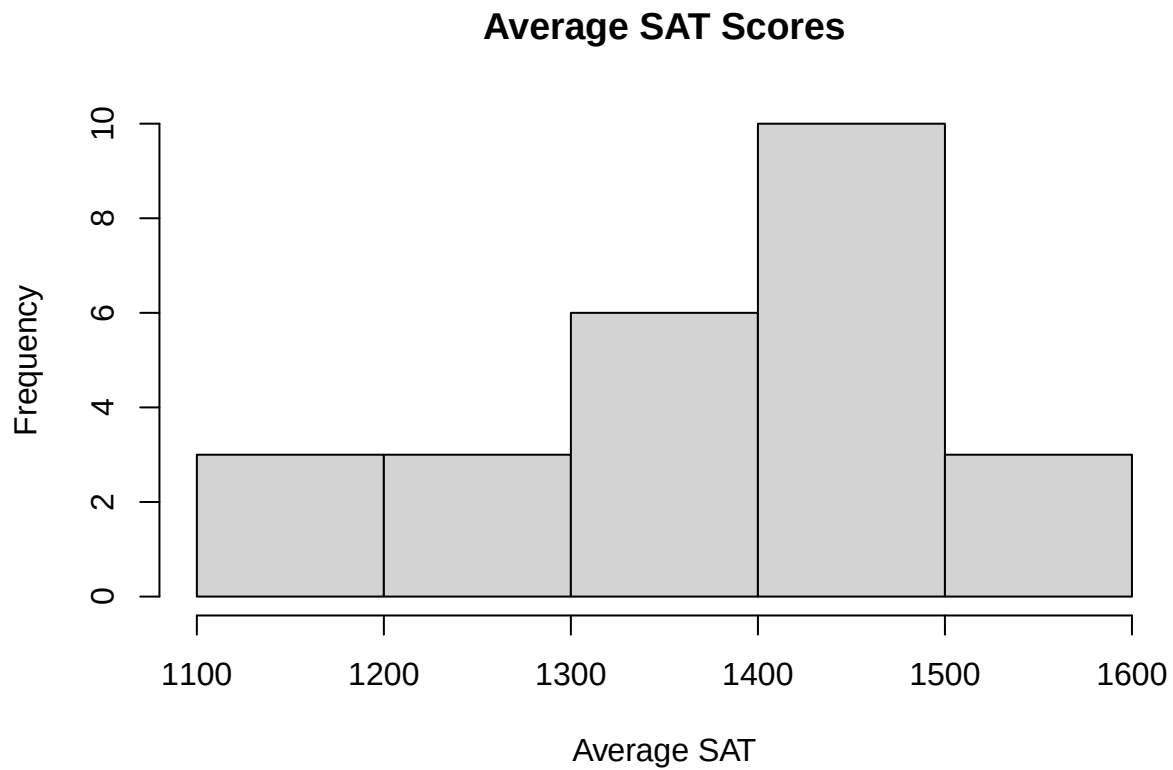
```
##
## Attaching package: 'caret'
##
##
## The following objects are masked from 'package:yardstick':
##
##   precision, recall, sensitivity, specificity
##
##
## The following object is masked from 'package:purrr':
##
##   lift
##
##
## Attaching package: 'maps'
##
##
## The following object is masked from 'package:purrr':
##
##   map
```

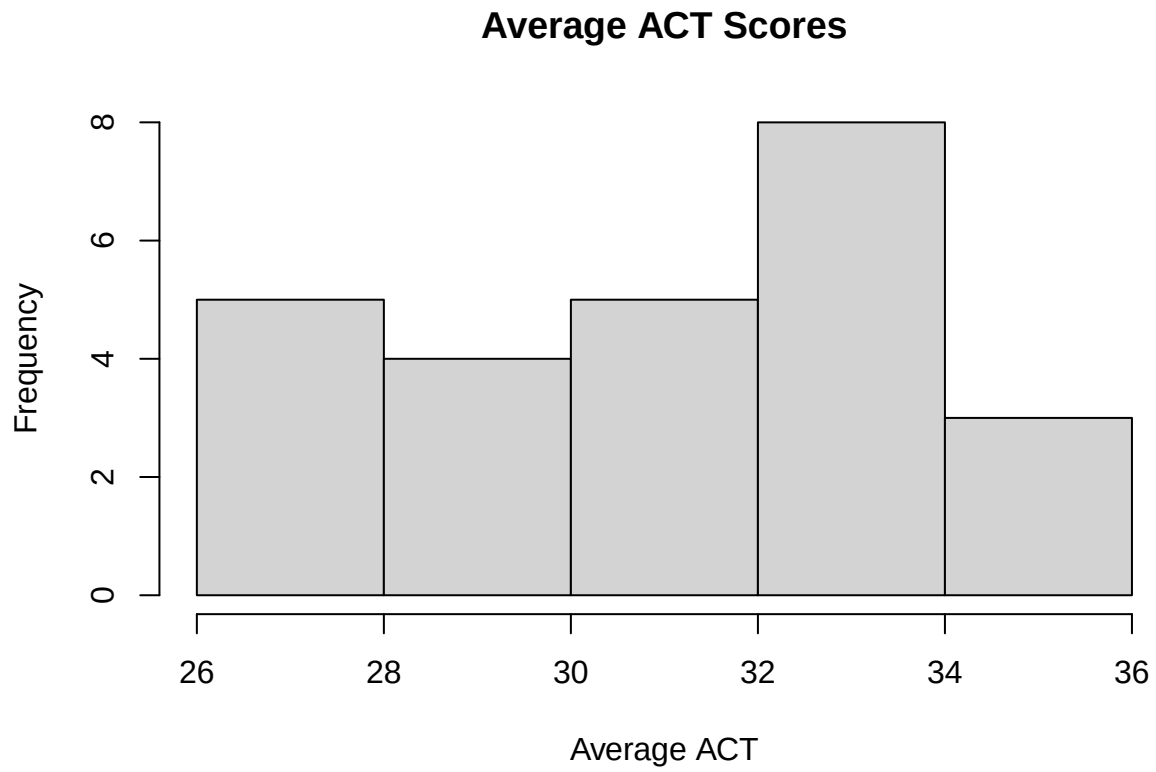
I'm being honest right now, I genuinely have not a clue what I am doing with this. But here we go! Special thanks to my man Lucas Qu and Varun Alluri for helping me figure out how to do the funky graphs and everything. Data is sourced for 25 colleges directly from the Google.com information bubble and edurank.com. Also thanks to ChatGPT for rescuing me when the graphs suddenly stopped working.

The data collected are as follows: Acceptance Rate - Undergraduate acceptance rate Average SAT - Average SAT composite score Average ACT - Average ACT composite score Cost - Total of undergraduate tuition, living expenses, etc. Aid Receiving - Portion of students receiving financial aid Undergraduate population - Number of undergraduate students Graduate population - Number of graduate students Percent Asian - Percentage of students that are asian Graduation rate - 4-year undergraduate graduation rate Publications - Number of publications attributed to the institution Citations - Number of citations of publications attributed to the institution Total staff - Total staff, including professors, administrators, etc. Private/Public - If a school is public or private Latitude - Latitude coordinate of the college Longitude - Longitude coordinate of the college

```
## # A tibble: 25 x 16
##   College      Acceptance.Rate Average.SAT Average.ACT Cost Aid.Receiving
##   <chr>          <dbl>         <int>         <int> <int>      <dbl>
## 1 "Harvard"      0.03          1535          35 83538      0.72
## 2 "MIT"          0.04          1555          36 79850      0.79
## 3 "Boston Universi~ 0.14          1425          33 82760      0.53
## 4 "Tufts "       0.1           1500          34 84600      0.46
## 5 "Massachusetts -- 0.64          1370          31 33475      0.91
## 6 "Northeastern"   0.07          1495          34 81422      0.85
## 7 "Boston College" 0.17          1488          34 83746      0.53
## 8 "Brandeis"       0.39          1450          33 81914      0.68
## 9 "Massachusetts -- 0.81          1190          27 34750      0.99
## 10 "WPI"          0.57          1400          30 76040      0.97
## # i 15 more rows
## # i 10 more variables: Undergraduates <int>, Graduates <int>, Asian <dbl>,
## #   Graduation.Rate <dbl>, Publications <int>, Citations <int>,
## #   Total.Staff <int>, Private <int>, Latitude <dbl>, Longitude <dbl>
```

Histogram of colleges by ACT and SAT scores. Guess which ones are MIT and Harvard.



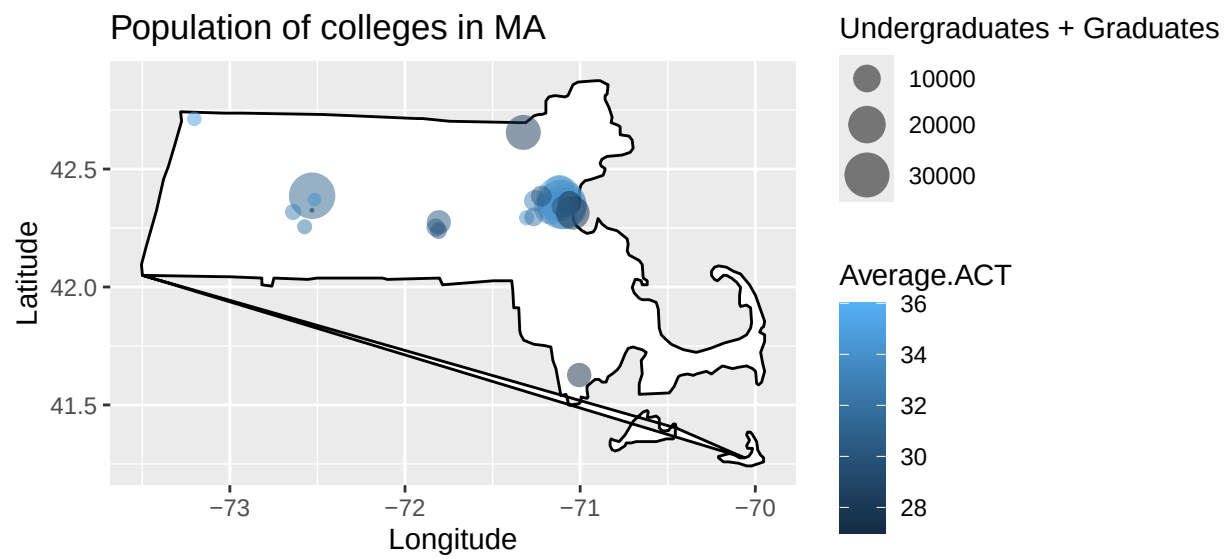


Geographical map of the 25 colleges, labeled, and color coded for private/public

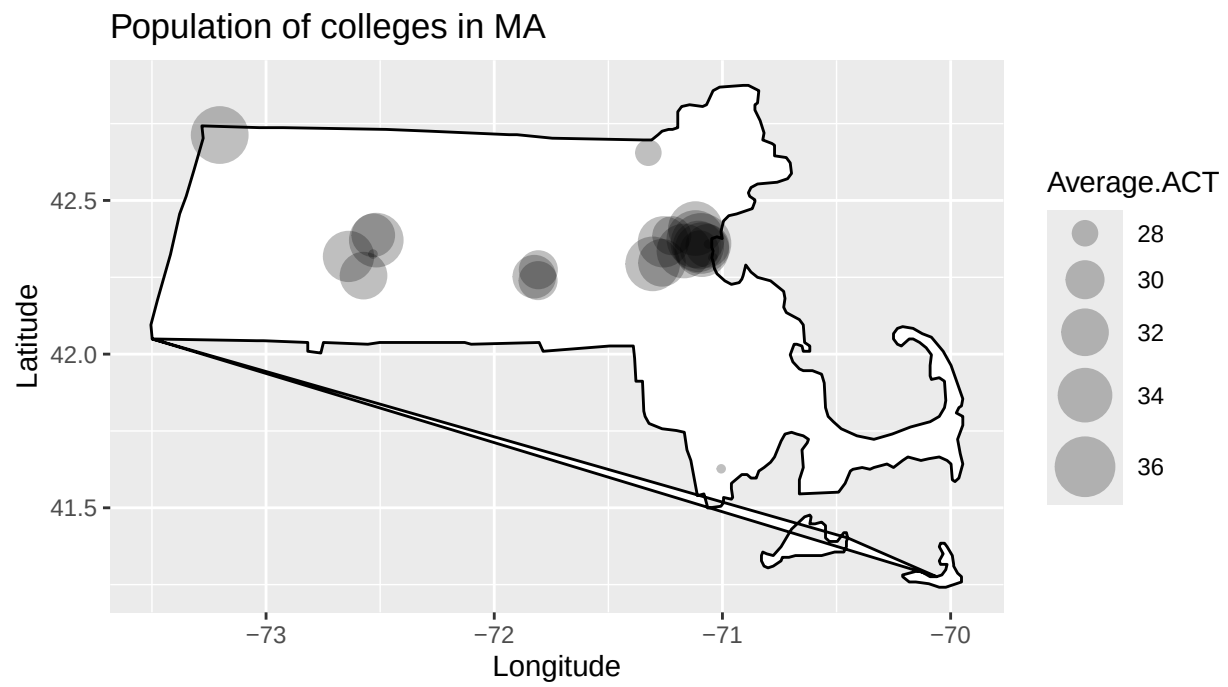


Geographical map of the 25 colleges, with the size of each dot proportional to the total student population and color equal to the average ACT of the school.

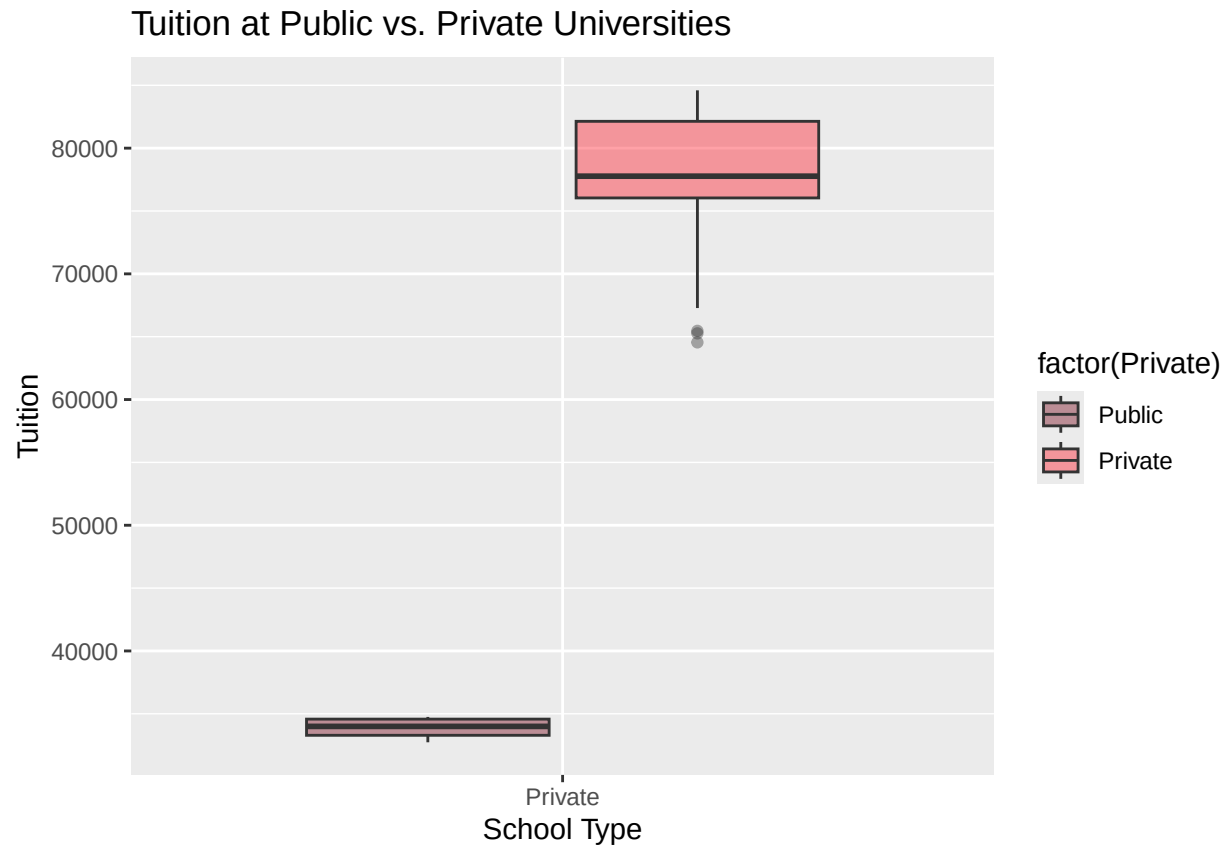
```
## Warning: The '<scale>' argument of 'guides()' cannot be 'FALSE'. Use "none" instead as
## of ggplot2 3.3.4.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



Just an easier view, bigger circle = higher ACT



Wow! Private schools are expensive!!!



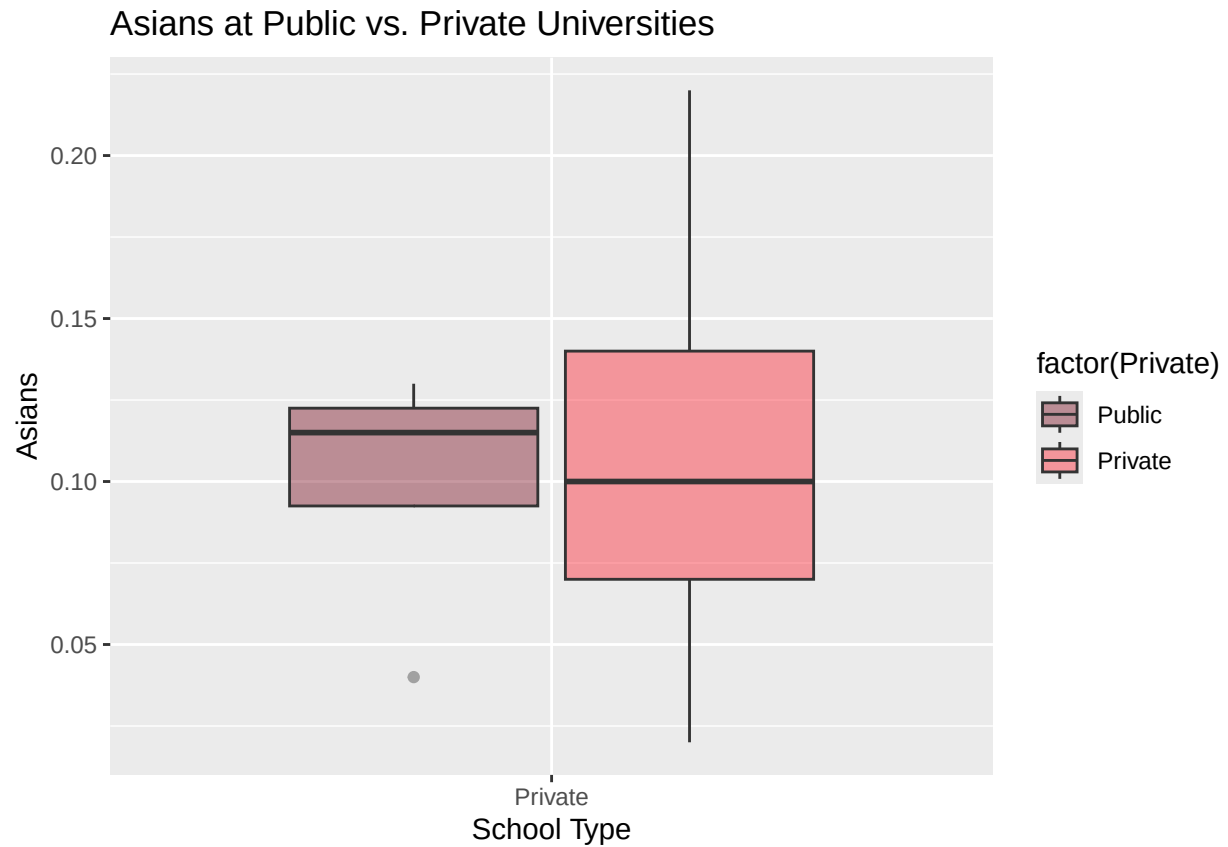
```
## [1] "Private Schools: "
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  64548  76040   77770   77229  82134   84600
```

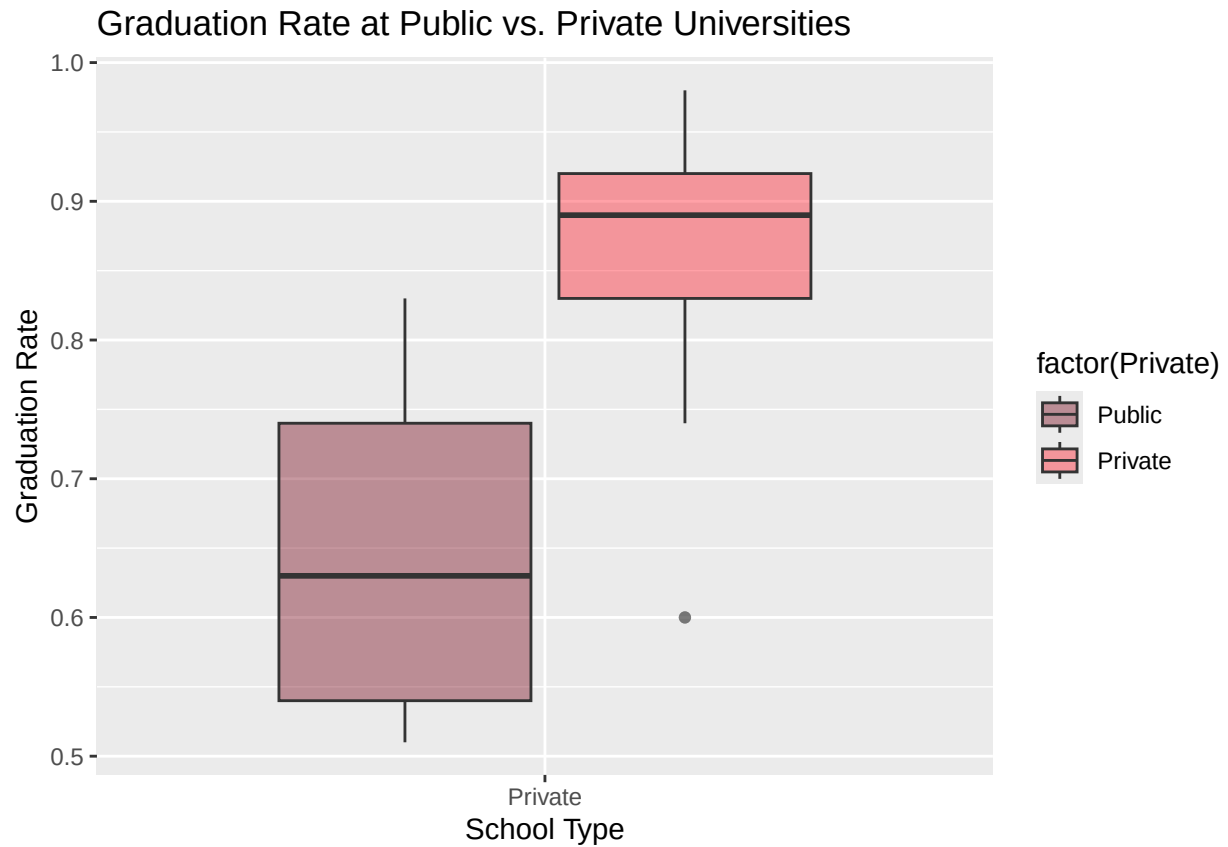
```
## [1] "Public Schools: "
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   32732   33289   33997   33869   34577   34750
```

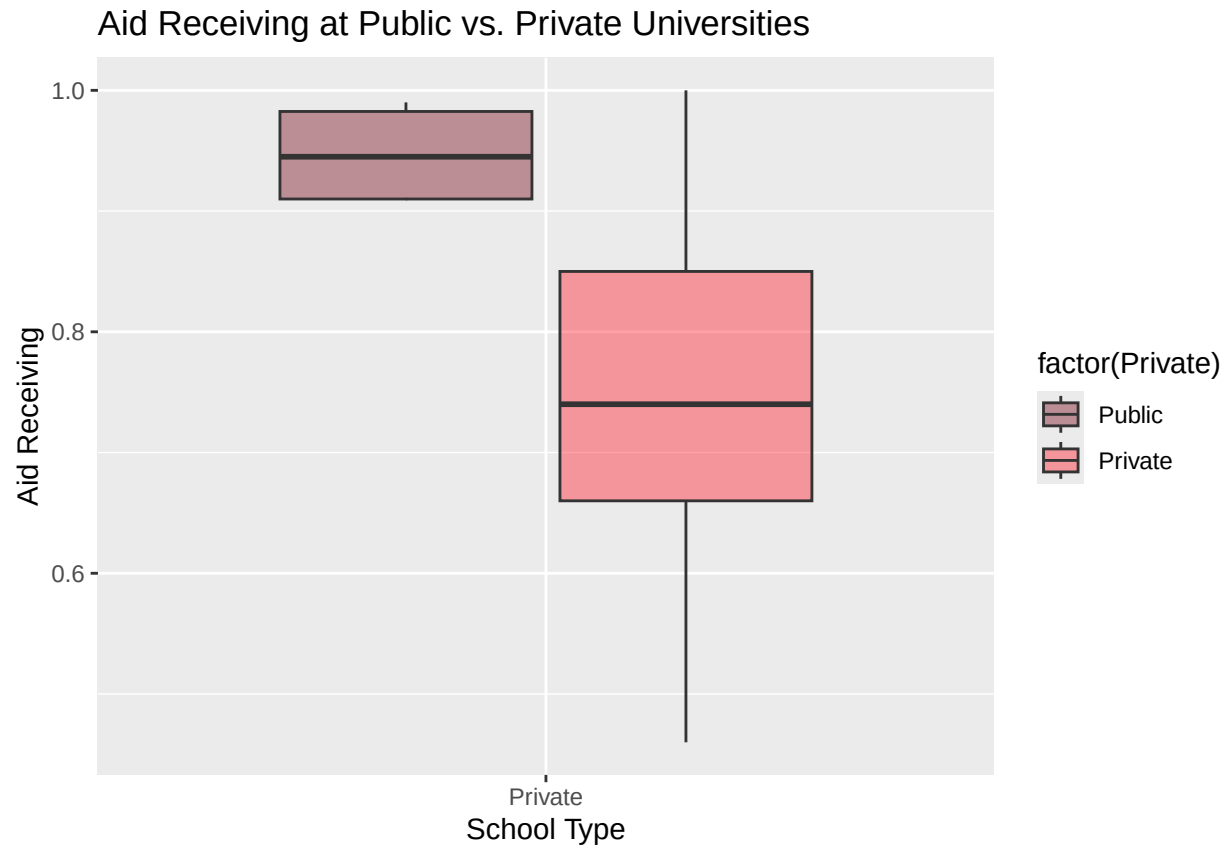
Comparison of asian population sizes between private and public universities. Much bigger range for publics, due to Hampshire on the low end, and Wellsley and MIT on the upper end.



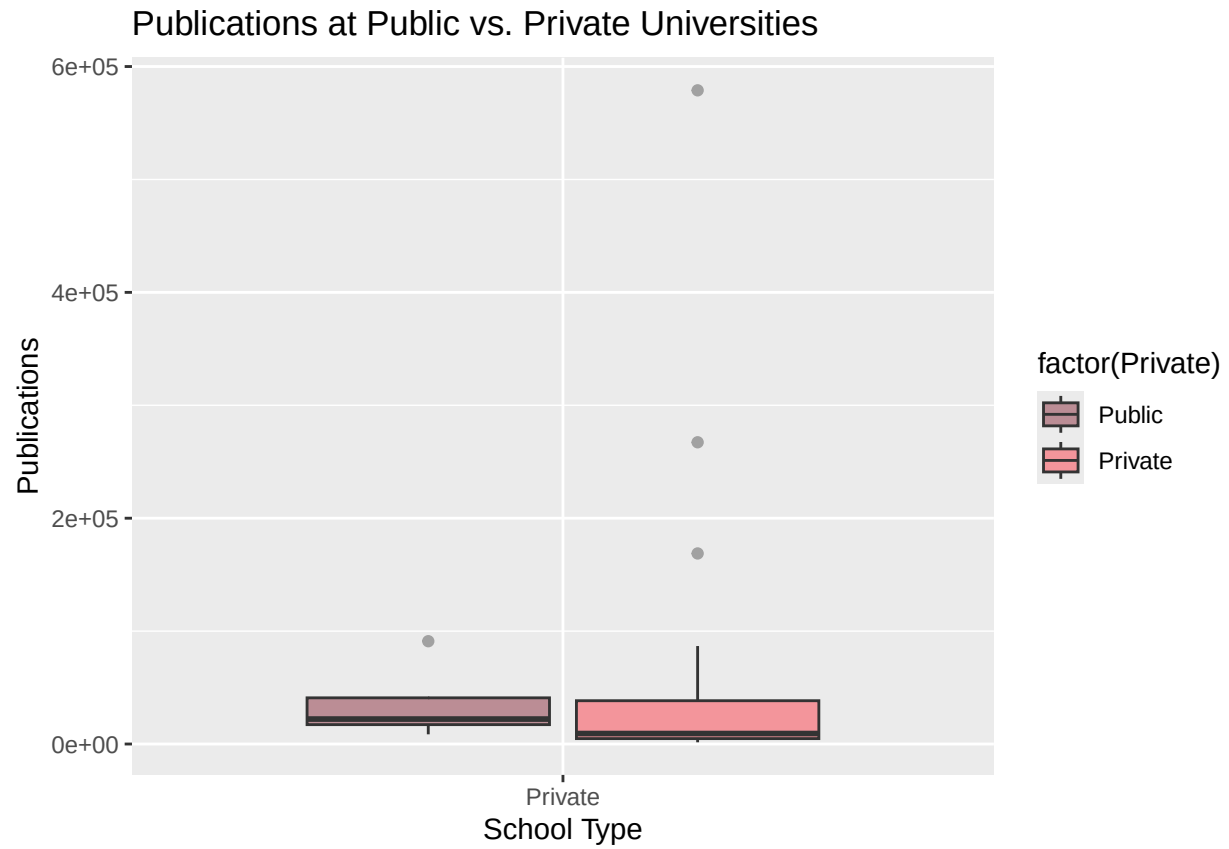
Graduation Rate of publics vs. privates. Likely higher at privates because they cost more, and thus the sunk cost fallacy kicks in.



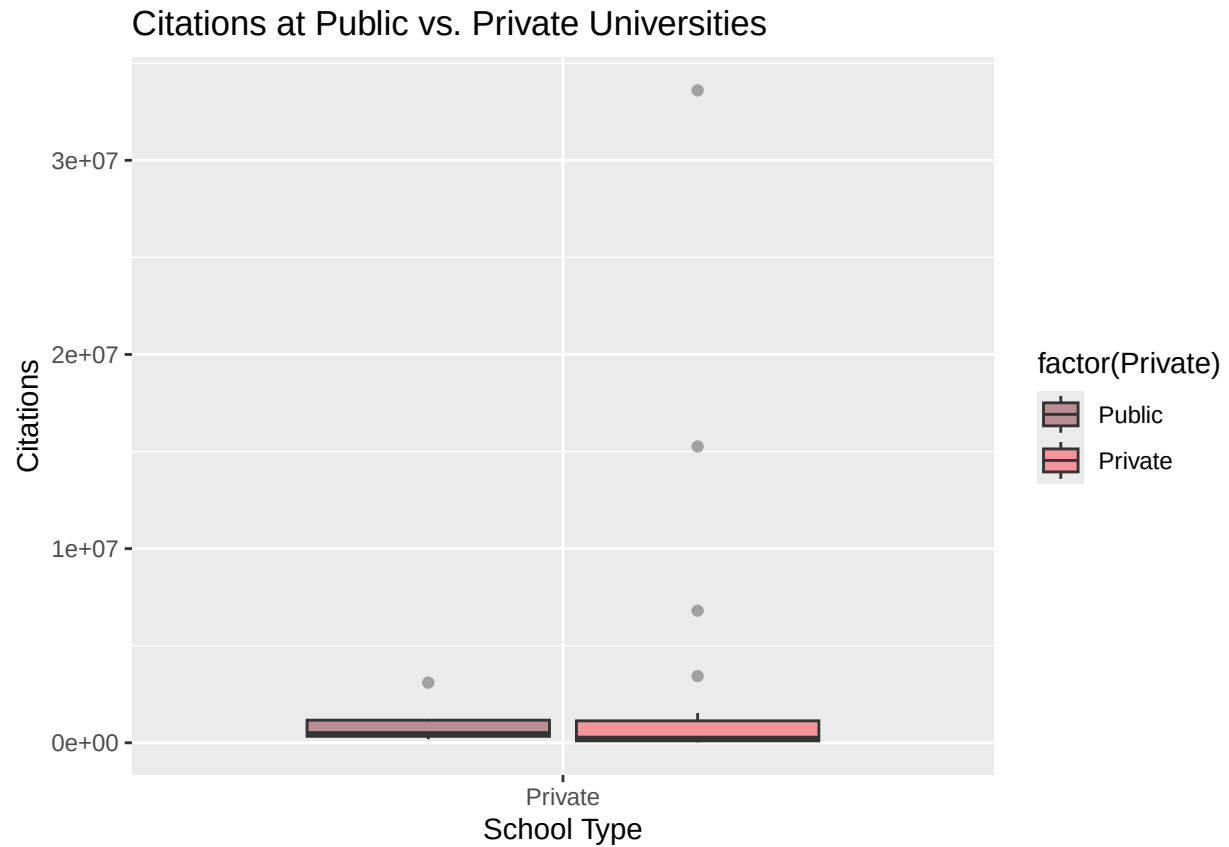
Publics give really good aid. Shocking. What is surprising is Hampshire University, which gives aid for all students, thus giving the Privates a much greater range.



Publics are much larger on average, and thus will have more publications. Can you guess what the four outliers are? (It's Harvard, MIT and BU for the privates, and UMass Amherst for the publics)

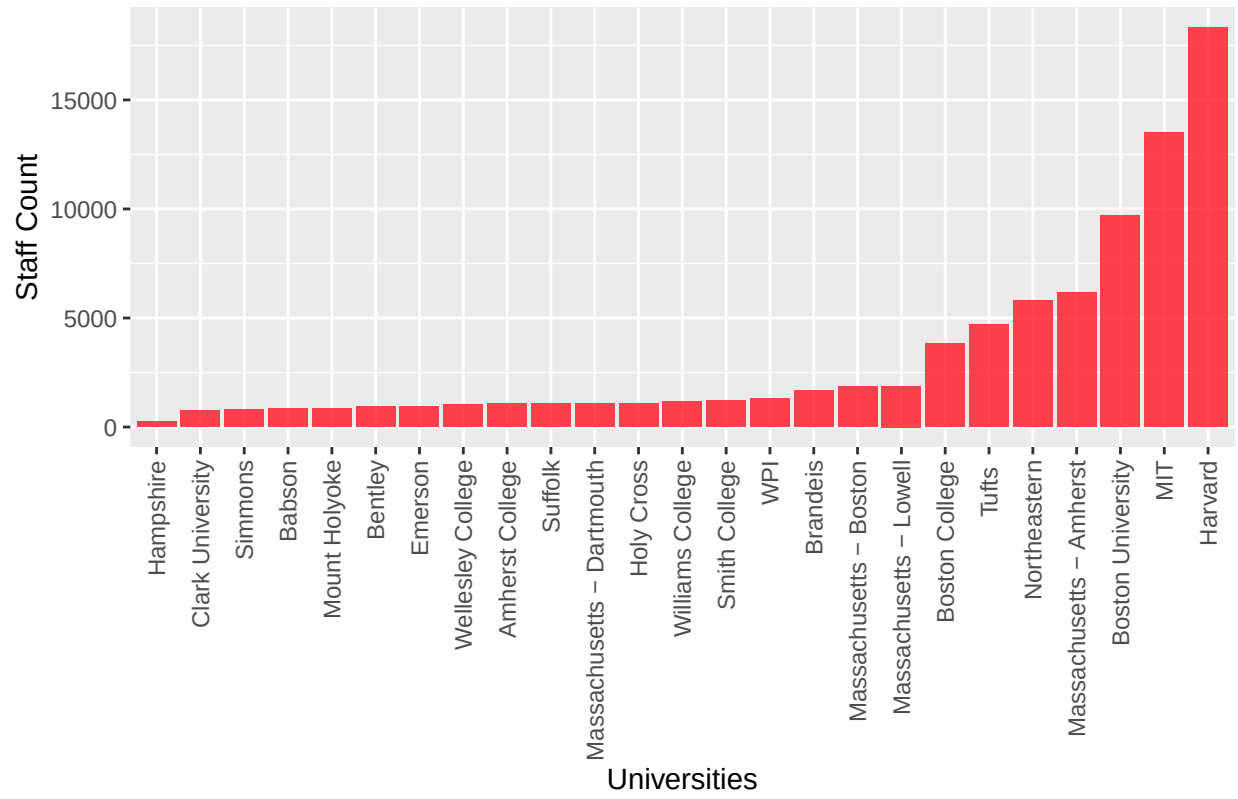


Same story as the last graph. Tufts is now also an outlier.

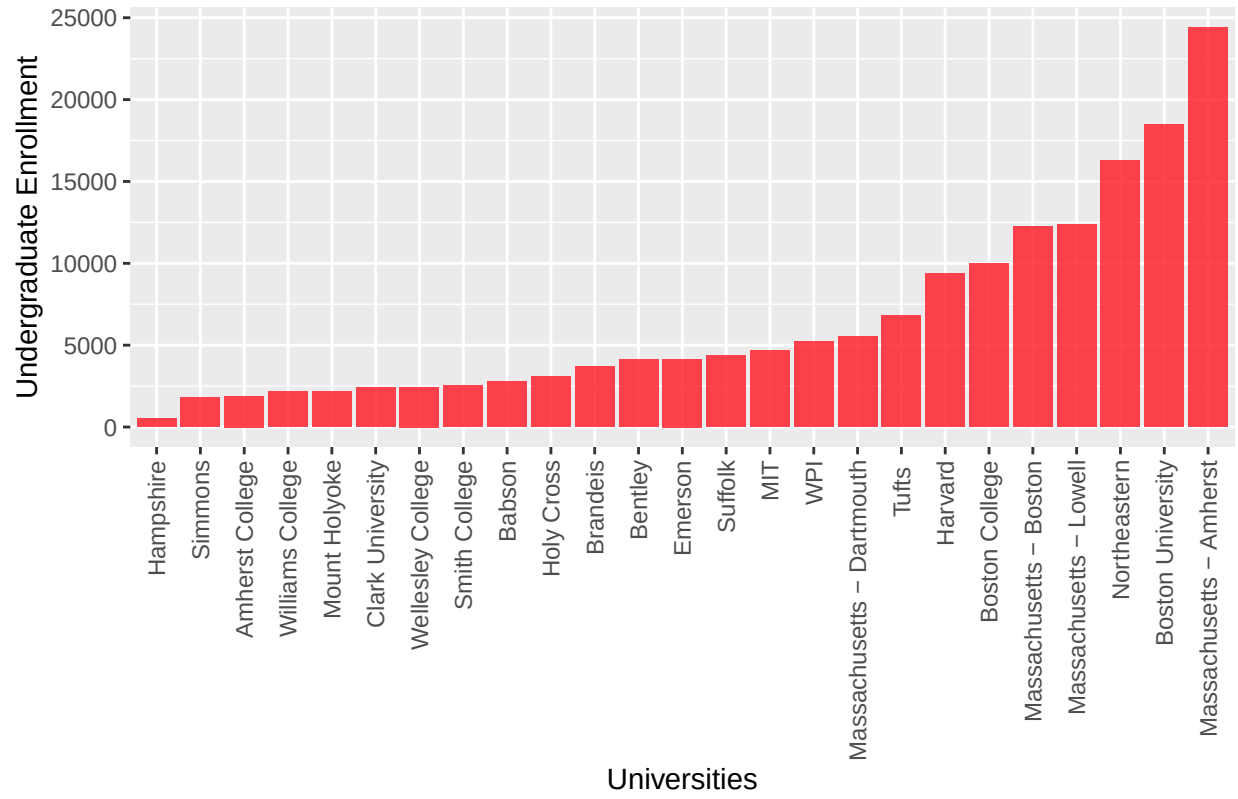


Another Lucas inspired graph, but this time I did a little thingy to sort the colleges by undergraduate student body size, then faculty count, then faculty/student ratio and student/faculty ratio. Look at MIT! More faculty than total students! Wow!

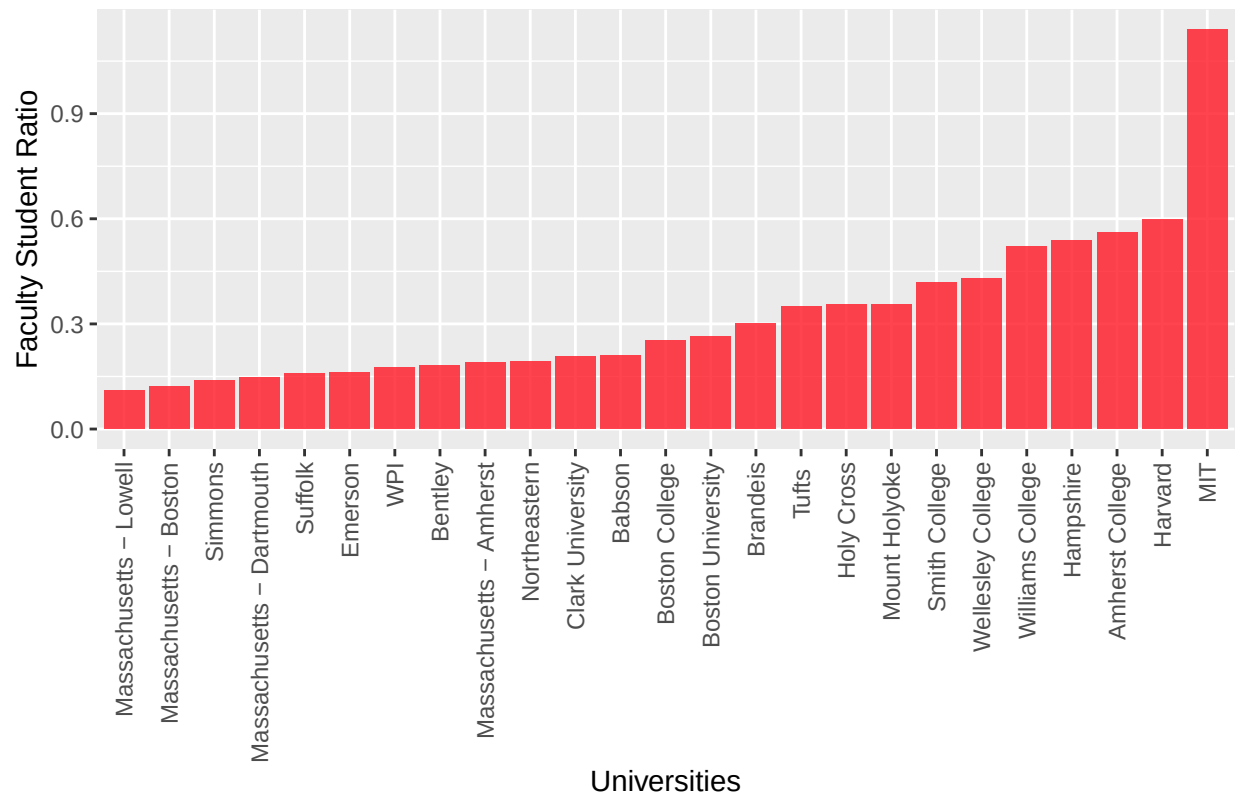
Staff Size of Universities in MA

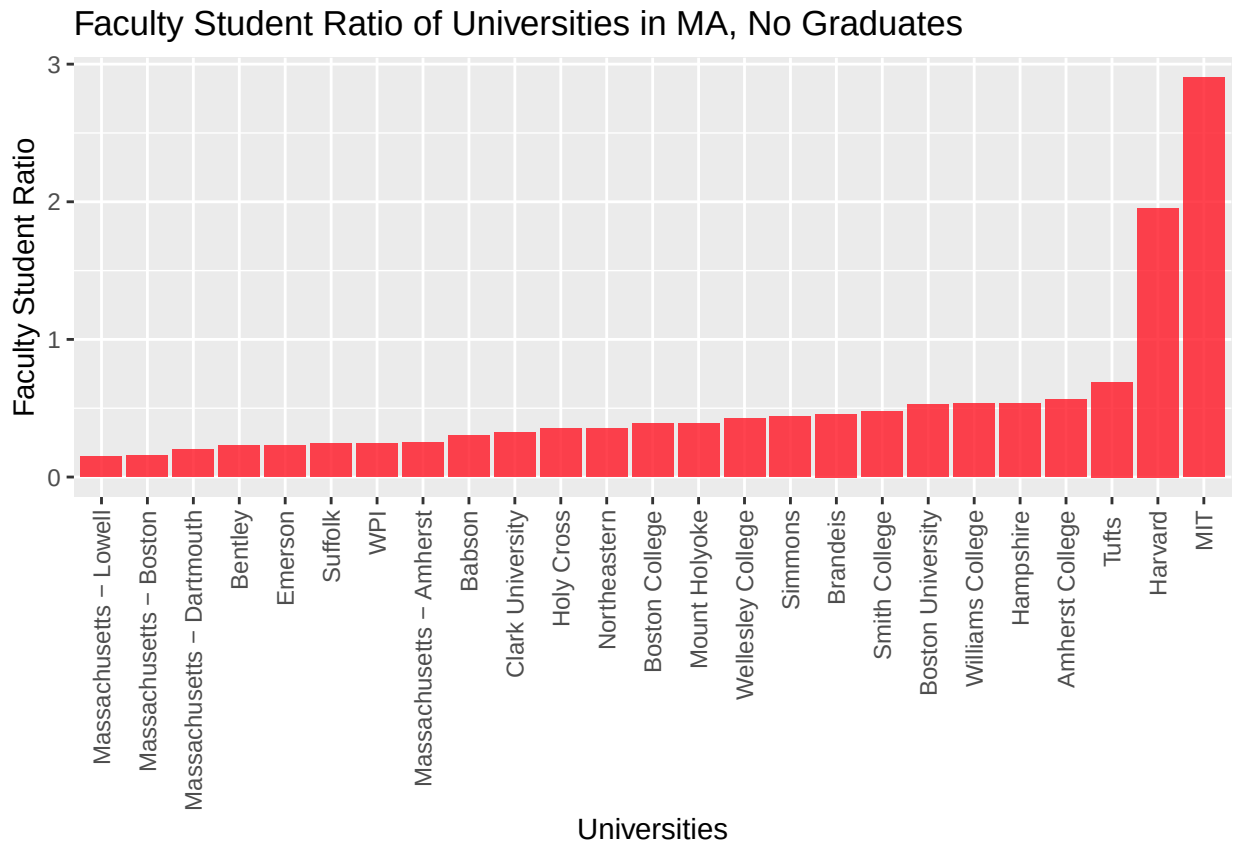


Undergraduate Student Body Size of Universities in MA

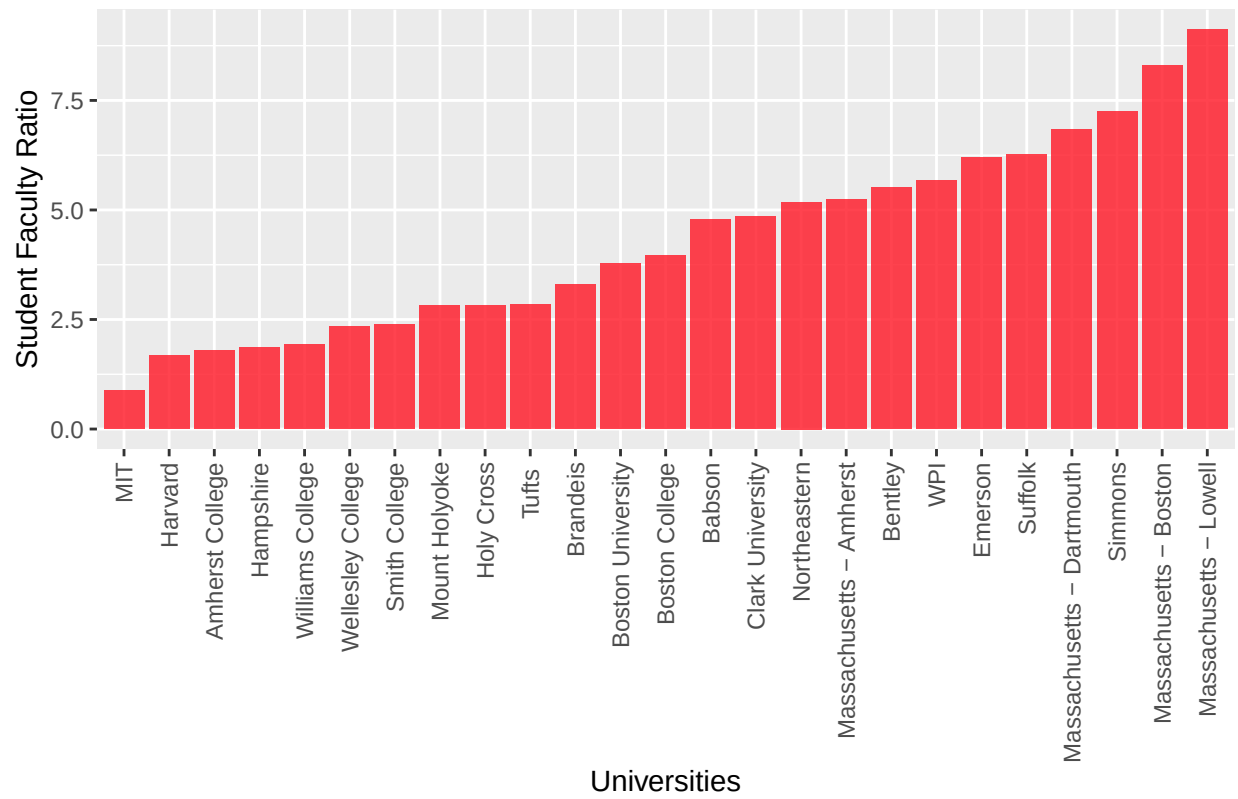


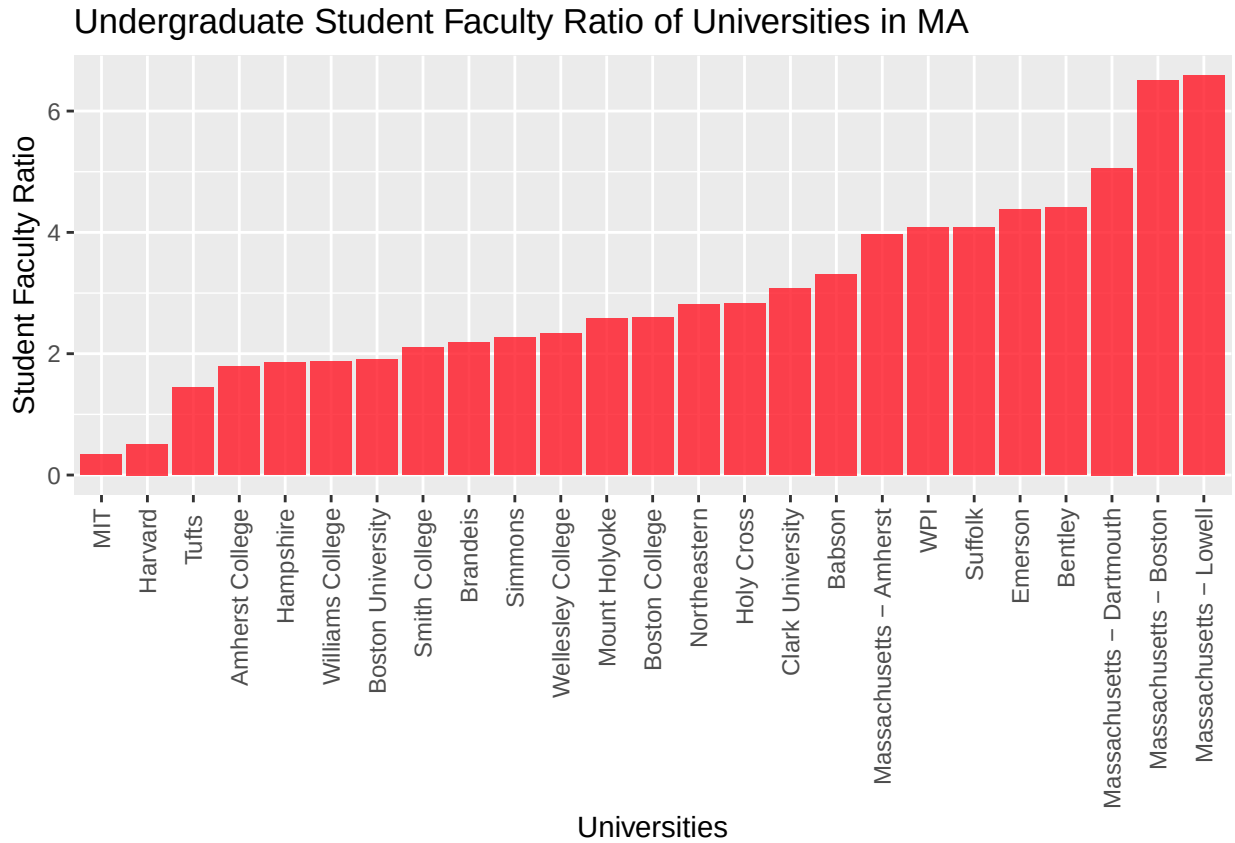
Faculty Student Ratio of Universities in MA





Student Faculty Ratio of Universities in MA





So closing thoughts. Pretty cool. It's interesting to see all the colleges bunched around the Boston and Cambridge area. Also the statistics are fun to look at too because MIT and Harvard literally skew everything. MIT especially, since they literally have more faculty than students!